

Canonical Spectral Model of ζ Zeros via the Persistence Dynamics of PT

Yan Senez
yan.senez@protonmail.com

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Abstract

We construct a canonical spectral model of the residual factor $R(s) = \zeta(s)/(\zeta_+(s)\zeta_-(s))$ of the Riemann zeta function, starting from the persistence dynamics of PT. Three main results are established.

1. A closed-form expression for the spectral amplitudes $\kappa_p(s) = 1 - (1 - p^{-s}) \exp(A_p p^{-s})$ such that $\det(I - D_R(s))^{-1} = R(s)$, verified to 10^{-12} in $\text{Re}(s) > 1$; D_R is the associated diagonal operator on $\ell^2(\mathcal{P})$.
2. An exact analytic trace formula $-d \log R/ds = \sum_p (\log p) [1/(p^s - 1) + A_p p^{-s}]$, verified to 10^{-26} , which reads as a Berry–Keating trace formula on the orbit lengths $\ell_p = \log p$.
3. A regularised operator trace $T(s) = \text{Tr}_{\text{reg}}[(s - i H_{\text{PT-BK}})^{-1} D_R(s)]$ whose distributional poles on the critical line are identified with the non-trivial zeros of $R(s)$. On $t \in [10, 200]$, the raw formula captures 74 zeros out of 79 at $|\Delta| < 1.5$ (on $t \in [10, 70]$: 17/17), with saturation beyond $P_{\text{max}} = 3000$. The explicit regularisation R_1 via Gaussian smearing ($\varepsilon = 0.2$) substantially improves precision: **75/79 zeros at $|\Delta| < 0.5$ and 47/79 at $|\Delta| < 0.1$** on the same domain, with a median $|\Delta| = 0.080$ against 0.52 for the raw formula. The test extended to $t \in [10, 500]$ (269 zeros) confirms the robustness: **267/269 at $|\Delta| < 1.5$ (99.3%), 242/269 at $|\Delta| < 0.5$ (90.0%), 139/269 at $|\Delta| < 0.1$ (51.7%)**, with capture by bands of 100 units that stays $\geq 86.6\%$ at $|\Delta| < 0.5$ across the five sub-bands — the performance does not collapse beyond $T = 200$.

The localisation $\text{Re}(s) = 1/2$ admits four equivalent readings internal to PT: symplectic Planck cell $u_* p_* = 2\pi$, multiplicative Haar Jacobian $du = dp/p$, unitary transformation $e^{su/2}$, and antiperiodic boundary condition forced by the involution T_3 of the sieve mod 3. The numerical residue $1/8$ separating the PT semi-classical density from the Riemann–von Mangoldt density admits a cohomological overdetermination: $1/8 = s^2/2 = \lambda_H$ (Higgs self-coupling), $1/8 = c_1/N_{\text{corners}}$ (first Chern class of the q_+/q_- spinor bundle), $1/8 = s/4$ (Berry phase per corner of the double barrier). The identity $s^2/2 = s/4$ has only $s = 0$ and $s = 1/2$ as solutions; the value $s = 0$ being excluded by **T1** (PT axiom), the consistency of the two readings overdetermines $s = 1/2$ without deriving it independently.

A structurally induced prediction — $\Delta_N(L \bmod q) = 1/(8q)$ for primitive Dirichlet L -functions — has been tested on the pre-computed zeros of LMFDB ($q \in \{3, 5, 7, 11\}$, ~ 140 zeros per character, $T \leq 200$). The test rejects the prediction at $\chi^2(\text{PT}) = 7.49$ against $\chi^2(H_0) = 0.023$ on four degrees of freedom. The calibration on the 50 first zeros of ζ shows that the residue $1/8$ is an identity between asymptotic forms (between the BK semi-classical density and the Riemann–von Mangoldt density), not a measurable shift in the count of zeros. The falsifiability discipline of the programme is reinforced and its validity domain made precise: the PT mechanism described here is specific to ζ , not to the Dirichlet family.

The article strictly distinguishes *understanding* the structure of the zeros and *proving* their localisation: the programme reaches the first objective at the operator level; the second remains an open problem in the classical sense, equivalent to the strict analytic continuation of $R(s)$ outside the half-plane of convergence.

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1 Introduction

The Riemann hypothesis asserts that the non-trivial zeros of the function $\zeta(s) = \sum_{n \geq 1} n^{-s}$ have real part $1/2$. Beyond its arithmetic formulation, this conjecture is also a question about the spectral nature of a certain operator, whose existence has been hypothesised since Hilbert and Pólya. Berry and Keating [1] in 1999, then Connes [2] in 1996, made the content of this spectral hypothesis precise: it is the dilation operator $H = (xp + px)/2$ quantised on the half-line, whose microcanonical semi-classical quantisation reproduces the Riemann–von Mangoldt density $N(\gamma) \sim (\gamma/2\pi) \log(\gamma/2\pi e)$ of the zeros.

The programme presented here does not seek a proof of the Riemann hypothesis. It seeks to reconstruct the Hilbert–Pólya–Berry–Keating (HP-BK) operator framework canonically from a single internal structure, that of the Theory of Persistence (PT), which derives the Standard Model from a unique arithmetic parameter $s = 1/2$ (theorem **T1**, [3]). The work revisits the RH question from three complementary angles.

First angle: analytic. We exhibit a factorisation $\zeta(s) = R(s) \zeta_+(s) \zeta_-(s)$, where $\zeta_{\pm}$ are defined by the amplitudes a_p, b_p coming from the q_+/q_- bifurcation of PT. The residual factor $R(s)$ admits an exact Fredholm representation $\det(I - D_R(s))^{-1} = R(s)$ with D_R diagonal and eigen-amplitudes $\kappa_p(s)$ in closed form (theorem 3.2 in §3.2). This representation is proved in $\text{Re}(s) > 1$ and verified numerically to 10^{-12} .

Second angle: operator-theoretic. We identify the canonical operator $H_{\text{PT-BK}} = (u p_u + p_u u)/2$ on the Mellin coordinate $u = \log p$ with its dynamical cut-off $u_{\max}(\gamma) = \gamma/\sqrt{2\pi}$. This cut-off is not postulated; it is forced by the symplectic Planck cell $u_{\star} p_{\star} = 2\pi$ of the canonical phase space (§4.1). We prove that the microcanonical semi-classical quantisation of this operator, in symmetric Weyl ordering with double barrier, reproduces the exact functional form of the Riemann–von Mangoldt density, up to a constant residue $1/8$, verified to 10^{-15} on four decades $\gamma \in [14, 10^3]$ (§4.1).

Third angle: spectral and distributional. The construction of a regularised trace $T(s) = \text{Tr}_{\text{reg}}[(s - iH_{\text{PT-BK}})^{-1} D_R(s)]$ links the canonical operator to the residual factor via the explicit trace formula

$$-\frac{d \log R}{ds} = \sum_p (\log p) \left[\frac{1}{p^s - 1} + A_p p^{-s} \right],$$

proved as an exact analytic identity in $\text{Re}(s) > 1$ (theorem 3.5) and distributionally continued to the critical line (§3.5). The Hadamard regularisation of this trace produces poles at the ordinates γ_n of the zeros of R ; we capture 17 zeros out of 17 tested on $t \in [10, 70]$ at $|\Delta| < 1.5$, and 74 out of 79 on $t \in [10, 200]$ (94%), with saturation beyond $P_{\max} = 3\,000$.

Intrinsic enrichment. The programme distinguishes itself from Berry–Keating–Connes by the intrinsic enrichment it provides. The apparently geometric numbers of the classical construction — the 2π of the Planck cell, the $1/8$ of the Maslov phase of the corners (semi-classical WKB correction of $-\pi/8$ per turning point at the corners of the microcanonical area domain, cf. [1, §3]) — receive a derivation proper to PT. The 2π is the canonical Liouville measure on (u, p_u) , that is the commutator $[u, p_u] = i$. The $1/8$ admits a triple cohomological reading:

$$\frac{1}{8} = \frac{s^2}{2} = \lambda_H \quad (\text{Higgs self-coupling, [4]}),$$

$$\frac{1}{8} = \frac{c_1}{N_{\text{corners}}} \quad (\text{first Chern class of the } q_+/q_- \text{ Kähler–Fisher spinor bundle}),$$

$$\frac{1}{8} = \frac{s}{4} \quad (\text{Berry phase per corner}).$$

The consistency of the three readings reduces to the equation $s^2/2 = s/4$, whose only solutions are $s = 0$ and $s = 1/2$; axiom **T1** excludes $s = 0$, which overdetermines the PT parameter modulo **T1** assumed (and not as an independent derivation from **T1**).

Honest falsification: Dirichlet. A prediction stemming from this framework, the formula $\Delta_N(L \bmod q) = 1/(8q)$ for primitive Dirichlet L -functions, has been numerically tested on LMFDB zeros and falsified (§8). This falsification is explicitly documented. It leads to reformulating the status of the $1/8$ established in §4.1: it is an asymptotic identity between smooth forms, not a measurable residue in the count of zeros. The PT mechanism described here is specific to ζ , and not to the Dirichlet family via a naive factorisation by the Haar measure modulo q .

Programme position. The distinction maintained throughout the text is the following. We seek to understand the structure of the zeros, to provide a canonical framework without ad hoc parameter, to derive the known numerical coincidences (2π , $7/8$, critical line) from a single internal structure. We do not seek to prove that all zeros lie on the critical line: this ultimate lock (equivalent to RH in the strict sense) remains open and does not seem easier than the classical version.

2 The PT Framework

2.1 The Fundamental Parameter

The Theory of Persistence (PT) takes as unique arithmetic axiom the theorem **T1** [3]:

$$s = \frac{1}{2}.$$

This parameter plays three distinct roles in the PT corpus, which will be progressively identified in the course of the article:

- **arithmetic:** s is the symmetry parameter of the forbidden transition at the first prime $p = 3$ of the sieve (theorem **T1**, [3]);
- **geometric:** s is the threshold $\gamma_p > 1/2$ for activity of a prime in the cascade (theorem **T5**, [5]);
- **spectral:** s is the real part appearing in the Mellin transform on the primes via the Jacobian of the multiplicative Haar measure.

A central reading of the present work is that these three manifestations are operator-theoretically identifiable (§4).

2.2 Fixed Point, Active Primes, Bifurcation

The PT sieve admits a unique fixed point $\mu^* = 15$. The arithmetically active primes at this fixed point are the prime divisors $\{3, 5, 7\}$ of μ^* ; the prime $p = 2$ plays the distinct role of parity channel (cf. §6). At this fixed point, the persistence dynamics bifurcates into two conjugate branches q_+ and q_- characterised by

$$q_+ = \frac{13}{15}, \quad q_- = \exp\left(-\frac{1}{15}\right). \quad (1)$$

This bifurcation is the central object of the zeta programme. It enters all the eigen-amplitudes and all the cohomological phases encountered below.

2.3 Multiplicative Haar Measure and Mellin Coordinate

The persistence cascade singles out a single invariant measure on the half-line of primes: the multiplicative Haar measure

$$du = \frac{dp}{p}, \quad u = \log p. \quad (2)$$

This measure is invariant under dilation $p \mapsto \lambda p$, and it is the unique measure invariant under the RG flow of PT (theorem **T2**, double stochasticity of the matrix T_3 , [3]). The coordinate $u = \log p$ will be called the *Mellin coordinate* throughout the article; it carries the measure additively invariant under translation $u \mapsto u + c$.

2.4 Canonical PT Factorisation of ζ

For $\text{Re}(s) > 1$ we define, from the bifurcation amplitudes,

$$\delta_p^\pm = \frac{1 - q_\pm^p}{p}, \quad a_p = \delta_p^+ (2 - \delta_p^+), \quad b_p = \delta_p^- (2 - \delta_p^-), \quad (3)$$

then

$$\zeta_+(s) = \exp\left(\sum_p a_p p^{-s}\right), \quad \zeta_-(s) = \exp\left(\sum_p b_p p^{-s}\right), \quad (4)$$

and the **residual factor**

$$R(s) = \frac{\zeta(s)}{\zeta_+(s)\zeta_-(s)}. \quad (5)$$

We finally set $A_p = a_p + b_p$, which will be the effective PT amplitude per prime in the sequel. Asymptotically, $A_p \sim 4/p$ for $p \rightarrow \infty$.

Canonical writing in holonomy angles. The amplitudes a_p, b_p also write

$$a_p = \sin^2 \theta_p(q_+), \quad b_p = \sin^2 \theta_p(q_-), \quad (6)$$

that is

$$A_p = \sin^2 \theta_p(q_+) + \sin^2 \theta_p(q_-), \quad (7)$$

where θ_p denotes the holonomy angle of the cascade at prime p (monograph ch. 10, fine structure, [6]). The equivalence between the two writings comes from $1 - \delta_p^\pm = q_\pm^p/p$, that is

$$\delta_p^\pm (2 - \delta_p^\pm) = 1 - (1 - \delta_p^\pm)^2 = 1 - (q_\pm^p/p)^2 = \sin^2 \theta_p(q_\pm)$$

on setting $\cos(\theta_p, q_\pm) = q_\pm^p/p$.

The zeta programme seeks a canonical spectral model of R , under the **conjectural hypothesis** of non-vanishing of $\zeta_+ \zeta_-$ (verified numerically on the tested zones) which identifies the non-trivial zeros of R with those of ζ . This hypothesis will be recalled at each critical step of the identification.

3 Spectral Model of the Zeros

This section presents the analytic and operator-theoretic framework. Four main results: closed-form expression of the amplitudes (κ_p), exact Fredholm representation of R , canalized operator for the intra-cascade coupling, and explicit trace formula.

3.1 Closed-Form Expression of the Amplitudes $\kappa_p(s)$

Definition 3.1. For each prime p and each $s \in \mathbb{C}$, we set

$$\boxed{\kappa_p(s) = 1 - (1 - p^{-s}) \exp(A_p p^{-s})} \quad (8)$$

By construction,

$$1 - \kappa_p(s) = (1 - p^{-s}) \exp(A_p p^{-s}),$$

and the asymptotic expansion for $p \rightarrow \infty$ gives

$$\kappa_p(s) = (1 - A_p) p^{-s} + O(p^{-2s}). \quad (9)$$

Interpretation. The fundamental scalar of the cascade is neither p^{-s} (Euler term) nor $\exp(A_p p^{-s})$ (bifurcated compensation), but their gap κ_p . We have the reading

$$1 - \kappa_p = \text{residual survival}, \quad \kappa_p = \text{spectral amplitude of the residue.}$$

3.2 Exact Fredholm Representation

Let $D_R(s)$ be the diagonal operator on $\ell^2(\mathcal{P})$ ($\mathcal{P}$ the set of primes) defined by $D_R(s) e_p = \kappa_p(s) e_p$.

Theorem 3.2 (Fredholm representation of R). *For $\text{Re}(s) > 1$, the operator $D_R(s)$ is trace class, and*

$$\boxed{\det(I - D_R(s))^{-1} = R(s).} \quad (10)$$

Proof. For $\text{Re}(s) > 1$, $\sum_p |\kappa_p(s)| < \infty$ by the asymptotic expansion (9); the diagonal operator is therefore trace class. Direct computation gives

$$\begin{aligned} \det(I - D_R(s))^{-1} &= \prod_p (1 - \kappa_p)^{-1} = \prod_p (1 - p^{-s})^{-1} \exp(-A_p p^{-s}) \\ &= \zeta(s) \exp\left(-\sum_p A_p p^{-s}\right) = R(s), \end{aligned}$$

using the classical Euler product for ζ and the definition of R in (5). $\square$

Scope of the representation. The proof is direct algebraic: (10) is a *determinantal rewriting* of the definition of R . Its value is to provide the *spectral target* that any candidate canonical PT operator must reproduce; it is not an independent argument for the spectral structure of R .

Numerical verification. For $s = 1.3 + 14.134725i$, $P = 10^6$ truncated primes, the residual error between $R(s)$ computed by the Fredholm product (10) and $R(s)$ computed from ζ via `mpmath` is of the order of $2.0 \cdot 10^{-12}$. At $\sigma = 1.1$ the precision drops to $\sim 1.3 \cdot 10^{-3}$, reflecting the approach to the boundary of absolute convergence, not a defect of the representation.

Remark 3.3 (Non-vanishing hypothesis). The identification $\text{zeros}(R) = \text{zeros}(\zeta)$ used in the sequel assumes $\zeta_+(s) \zeta_-(s) \neq 0$ on the zones of interest (in particular the critical line). This hypothesis is verified numerically on the tested zones but is not proved; it is **conjectural** and must be treated as such (cf. §9).

3.3 Canalized Operator for the Coupling $\sum_p A_p p^{-s}$

To model the intra-cascade coupling (the sum $\sum_p A_p p^{-s}$) as the spectrum of a canonical operator *without adjustable parameter*, we construct

$$\mathcal{H}_{\text{PT}} = L^2(\Sigma_{\text{pers}}, du) \otimes L^2(\widehat{\mathbb{Z}/M\mathbb{Z}}). \quad (11)$$

The first component is the continuous Mellin coordinate on Σ_{pers} (persistence surface, of genus 14 in the PT corpus [7]); the second is discrete, on the characters of the multiplicative group modulo M (cascade by Chinese remainder theorem, M primorial product). The inter-window coupling is constructed by regularised Bergman kernel

$$B_{\text{BF}}(z_i, z_j) = \frac{\varepsilon_i \varepsilon_j}{|z_i - z_j|^2 + \varepsilon_i \varepsilon_j}, \quad \varepsilon_i = \frac{\varepsilon_*}{\rho_i}. \quad (12)$$

Lemma 3.4 (Geometric determination of ε_*). *The regularised Bergman covariance at maximal resolution ($\varepsilon_{\text{reg}} = 10^{-3}$, packets of individual prime size) on the atomic measure $\pi_{\log}$ of the Mellin coordinate gives*

$$\varepsilon_*^{\text{geom}} = 1.51. \quad (13)$$

Sketch. The regularised Bergman covariance of the kernel (12) on the atomic measure $\pi_{\log} = \sum_p \delta(u - \log p)$ is, at resolution ε_{reg} , the weighted average $\text{Cov}(\varepsilon_{\text{reg}}) = \iint B_{\text{BF}}(u, u')^2 \pi_{\log}(du) \pi_{\log}(du')$. The numerical sweep over values of ε_{reg} by steps $10^{-3}, 10^{-2}, 10^{-1}, 1$ gives a covariance respectively of 0.0252, 0.0794, 0.2408 and 0.680, that is a factor $\varepsilon_*^{\text{geom}} = 1.51$ at the maximal resolution (natural reference of the continuous model). This is the canonical value injected into the canalized operator. $\square$

Performance. With $\varepsilon_*^{\text{geom}}$ injected without adjustment, the median error on $\sum_p A_p p^{-s}$ is 0.085 at $M = 1155$, $W = 24$ — a factor ~ 85 above the adaptive smoothed scalar baseline (0.001, which uses a tuned β). The contribution is not absolute precision but the **absence of a free parameter**: (13) replaces an empirical tuning by a canonical geometric covariance.

3.4 Explicit Trace Formula

Theorem 3.5 (Explicit trace formula). *For all s with $\text{Re}(s) > 1$,*

$$\boxed{-\frac{d}{ds} \log R(s) = \sum_p \frac{\kappa'_p(s)}{1 - \kappa_p(s)} = \sum_p (\log p) \left[\frac{1}{p^s - 1} + A_p p^{-s} \right]}. \quad (14)$$

Proof. By logarithmic differentiation of $1 - \kappa_p(s) = (1 - p^{-s}) \exp(A_p p^{-s})$. Set $z = p^{-s}$, so that $dz/ds = -\log p \cdot z$. A direct computation gives

$$\frac{\kappa'_p(s)}{1 - \kappa_p(s)} = -(\log p) \left[A_p z - \frac{z}{1 - z} \right] = -(\log p) \left[A_p p^{-s} - \frac{1}{p^s - 1} \right].$$

Summing over p and changing the overall sign yields the announced formula. A *sanity check* via $-\zeta'/\zeta(s) = \sum_n \Lambda(n) n^{-s} = \sum_p (\log p)/(p^s - 1)$ gives the same expression for the first term. $\square$

Numerical verification. At $s = \sigma + 14.134725i$ with $\sigma \in \{1.2, 1.5, 2, 3\}$ and 2000 truncated primes, the discrepancy between the two forms of (14) is of the order of 10^{-26} (precision `mpmath` 25 digits). The identity is therefore exact in the algebraic sense.

Berry–Keating reading. The dilation operator $H_{\text{PT-BK}} = (u p_u + p_u u)/2 = -i(u \partial_u + 1/2)$ generates the flow $e^{-itH_{\text{PT-BK}}} f(u) = e^{-t/2} f(e^{-t}u)$. On the atomic measure $\nu = \sum_p \delta(u - \log p)$, the periodic orbits of the flow starting from $\log p_0$ have lengths $\ell_p = k \log p_0$, $k \geq 1$, corresponding to the powers p_0^k . The term $(\log p)/(p^s - 1) = \sum_{k \geq 1} (\log p) p^{-ks}$ identifies then with the sum over these orbits, and $A_p p^{-s}$ represents the q_+/q_- holonomic coupling contribution per prime.

3.5 Regularised Operator Trace

The compact writing of the trace formula (14) in terms of the resolvent of $H_{\text{PT-BK}}$ and of the operator D_R uses an atomic Mellin projection

$$\Pi : L^2([u_{\min}, u_{\max}], du) \rightarrow \ell^2(\mathcal{P}_\gamma), \quad \Pi(f)_p = f(\log p) \sqrt{\log p}, \quad (15)$$

where $\mathcal{P}_\gamma = \{p \text{ prime} : u_{\min} \leq \log p \leq u_{\max}(\gamma)\}$. The factor $\sqrt{\log p}$ is the square root of the atomic multiplicative Haar measure. The regularised trace then writes

$$T(s) := \text{Tr}_{\text{reg}} \left[(s - iH_{\text{PT-BK}})^{-1} D_R(s) \right] = \sum_{p \in \mathcal{P}_\gamma} \kappa_p(s) (\log p) \left[\frac{1}{p^s - 1} + A_p p^{-s} \right]. \quad (16)$$

In $\text{Re}(s) > 1$, $T(s) \equiv -d \log R / ds$. On the critical line $\text{Re}(s) = 1/2$, the prime sum is not absolutely convergent; it is however distributionally convergent as an element of $\mathcal{S}'(\mathbb{R}_t)$, and the Hadamard regularisation gives a functional sense to $T(1/2 + it)$.

Three equivalent regularisations. We will use three regularisations interchangeable in the distributional sense.

(R_1) Smearing by test function. For ϕ_ε a compactly supported approximation of the identity with width ε , we set $I_\varepsilon(\gamma_0) := \int \phi_\varepsilon(t - \gamma_0) F(1/2 + it) dt$. The peaks $I_\varepsilon(\gamma_n)$ behave as c/ε when $\varepsilon \rightarrow 0^+$ (smearing regularisation).

(R_2) Abel-Hadamard summation. With weight $p^{-\varepsilon}$, $\varepsilon > 0$, $F_\varepsilon(t) := \sum_p (\log p) p^{-\varepsilon} [1/(p^{1/2+it} - 1) + A_p p^{-1/2-it}]$. By Fourier transform, F_ε is the convolution of F by the Gaussian ϕ_ε : R_2 is equivalent to R_1 modulo the choice of regularising kernel.

(R_3) Subtraction of the principal pole. $\tilde{F}(s) := F(s) - 1/(s - 1)$ isolates the simple pole at $s = 1$ and applies the Hadamard extraction to the remaining part.

The three regularisations coincide in the distributional sense at the poles of R . The strict functional proof of the analytic continuation of T to the critical line — beyond distributional convergence — remains open (§9).

Self-correction: G_HP4.d falsified. A first test posed the naive identity “direct spectrum of $H_{\text{PT-BK}}$ bounded by double barrier $= \{\gamma_n\}$ ”. This test was falsified: the direct spectrum is regular (Bohr–Sommerfeld asymptotically linear), while the γ_n are irregular, and the ratio between the two spectra diverges. This falsification motivated the move to the distributional regularised trace: the zeros of ζ are *distributional poles* of $T(s)$, not direct eigenvalues of $H_{\text{PT-BK}}$. The falsifiability discipline of the programme includes this correction on the same footing as the Dirichlet falsification of §8.

4 Localisation $\text{Re}(s) = 1/2$: Four Equivalent Mechanisms

The central object of this section is the PT-canonical explanation of the critical line. Four apparently distinct mechanisms appear at different stages of the programme. They are in fact four faces of a single structure: the invariant measure of the persistence dynamics.

4.1 Symplectic Planck Cell: $u_\star p_\star = 2\pi$

The dilation operator $H_{\text{PT-BK}}$ generates the scale transformations on the Mellin coordinate. Its canonical Liouville measure is $du dp_u/(2\pi)$, a direct consequence of the canonical commutator $[u, p_u] = i$.

Proposition 4.1 (PT Planck cell). *The unit cell of the canonical phase space is*

$$\boxed{u_\star p_\star = 2\pi} \quad (17)$$

with the symmetric choice $u_\star = p_\star = \sqrt{2\pi} \approx 2.5066$ (fixed point of the involution $(u, p_u) \leftrightarrow (p_u, u)$). At fixed energy γ , the hyperbola $H_{\text{PT-BK}} = \gamma$ enters the admissible space only on the segment

$$u \in [u_\star, \gamma/p_\star] = [\sqrt{2\pi}, \gamma/\sqrt{2\pi}],$$

and the canonical dynamical cut-off equals

$$u_{\max}(\gamma) = \frac{\gamma}{\sqrt{2\pi}}. \quad (18)$$

Proposition 4.2 (Microcanonical semi-classical density). *With the cut-off (18) and the double barrier imposed by the involution $(u, p_u) \leftrightarrow (p_u, u)$, the microcanonical quantisation of $H_{\text{PT-BK}}$ in symmetric Weyl order produces the integrated density*

$$N_{\text{PT}}(\gamma) = \frac{\gamma}{2\pi} \log \frac{\gamma}{2\pi e} + 1. \quad (19)$$

Sketch. The microcanonical area between the curve $H_{\text{PT-BK}} = \gamma$ and the barrier $u = u_\star$ in the quadrant $u, p_u > 0$ equals $A(\gamma) = \int_{u_\star}^{u_{\max}(\gamma)} \gamma/u du = \gamma \log(u_{\max}(\gamma)/u_\star) = \gamma \log(\gamma/2\pi)$. The double barrier (the mirror sector $u, p_u < 0$ obtained by the involution) doubles this area. The semi-classical Weyl quantisation in symmetric order gives $N_{\text{PT}}(\gamma) = A(\gamma)/\pi + \text{const}$. The constant is fixed to +1 by the double self-adjoint extension of $H_{\text{PT-BK}}$ at the boundaries (see §4.4). Direct algebraic computation gives (19). $\square$

Comparison with Riemann–von Mangoldt. The classical Riemann–von Mangoldt density [8] is

$$N_{\text{RvM}}(\gamma) = \frac{\gamma}{2\pi} \log \frac{\gamma}{2\pi e} + \frac{7}{8} + O(1/\gamma). \quad (20)$$

The functional form of (19) and (20) is identical. The constant gap

$$N_{\text{PT}}(\gamma) - N_{\text{RvM}}(\gamma) = \frac{1}{8} \quad (21)$$

is verified to 10^{-15} on four decades, $\gamma \in [14, 10^3]$. The 2π appearing in the density $N(\gamma)$ is the canonical Liouville measure on (u, p_u) , consequence of the commutator $[u, p_u] = i$; the shift $\text{Re}(s) = 1/2$ is derived from the Jacobian of the multiplicative Haar measure (§4.2). The residue $1/8$ admits the cohomological overdetermination of §5.

4.2 Multiplicative Haar Jacobian

For a stationary mode $f(\tau, u) = e^{i\omega\tau} \psi(u)$ of the d'Alembert operator on the Mellin coordinate (the Fisher metric of Σ_{pers} being Lorentzian for $\mu > \mu_c \approx 6.97$, [7]), the Mellin shift under the multiplicative Haar measure $d\pi_{\log} = \sum_p \delta_{\log p} d\log p$ produces a factor $1/p = e^{-u/2}$ in front of the transform. More precisely, the pole of the Mellin transform moves from $s = \omega$ to

$$\boxed{s = \frac{1}{2} + i\omega.} \quad (22)$$

This is the shift of the multiplicative Haar measure. The real part $1/2$ is not postulated: it comes from the natural Jacobian of the invariant Haar measure on the primes.

4.3 Unitary Transformation $e^{su/2}$

The Ruelle–PT self-adjointness theorem ([6]) establishes that the discrete Ruelle–PT operator L_s is self-adjoint for the Gibbs measure $\mu_s(p) = p^{-s}$ via the detailed balance associated with the antidiagonal symmetry $T_3 = T_3^\top$. In Mellin coordinate, the Gibbs measure becomes $\mu_s(p) dp = e^{-su} du$. The unitary transformation

$$U : f(u) \mapsto e^{-su/2} f(u) \quad (23)$$

brings the operator L_s on $L^2(\mathbb{R}, e^{-su} du)$ to $L^2(\mathbb{R}, du)$ by translating $H_{\text{PT-BK}}$ by $is/2$. The shift appears literally in the exponent: it is the operator identification between discrete detailed balance (**T2** of the sieve) and continuous self-adjointness of $H_{\text{PT-BK}}$.

4.4 Antiperiodic Boundary Condition

On $L^2([u_{\min}, u_{\max}], du)$ with $0 < u_{\min} < u_{\max} < \infty$, the defect indices of $H_{\text{PT-BK}}$ equal $(n_+, n_-) = (1, 1)$. By von Neumann, there exists a $U(1)$ family of self-adjoint extensions parameterised by

$$\psi(u_{\max}) = e^{i\theta} \psi(u_{\min}), \quad \theta \in [0, 2\pi). \quad (24)$$

Proposition 4.3 (PT-canonical selection: antiperiodicity $\theta = \pi$). *The involution*

$$\iota : u \mapsto u_{\max} + u_{\min} - u$$

*is the continuous projection of the antidiagonal $T_3 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ of the sieve mod 3. It is the unique continuous involution preserving the boundaries as orbit and is an isometry of L^2 . The boundary term $\iota H_{\text{PT-BK}} \iota^{-1} + H_{\text{PT-BK}}$ vanishes **only** for $\theta = \pi$ (antiperiodic), which forces the spectral symmetry $\{\gamma_n, -\gamma_n\}$.*

Sketch. The family (24) is parameterised by $\theta \in [0, 2\pi)$. The involution ι maps a self-adjoint extension with parameter θ to the one with parameter $-\theta$. Invariance under ι therefore imposes $\theta \equiv -\theta \pmod{2\pi}$, i.e. $\theta \in \{0, \pi\}$. The case $\theta = 0$ (periodicity) gives a symmetric spectrum but does not preserve the antisymmetry of $H_{\text{PT-BK}}$ at the boundaries; only $\theta = \pi$ satisfies $\iota H_{\text{PT-BK}} \iota^{-1} + H_{\text{PT-BK}} = 0$ and produces the required spectral symmetry $\{\gamma_n, -\gamma_n\}$. Verified numerically at $5 \cdot 10^{-12}$ (machine precision). $\square$

The continuous antiperiodicity is therefore the continuous image of the discrete antidiagonality T_3 of the sieve, which is itself the simplest expression of a non-trivial 2×2 stochastic matrix. Discrete detailed balance and continuous self-adjointness translate into each other through the unitary transformation (23).

4.5 Unification

The four mechanisms above manifest four facets of the same persistence dynamics, with its multiplicative Haar invariant measure and its **T2** reversibility (double stochasticity). The links $M1 \leftrightarrow M2$ (Liouville measure $\rightarrow$ multiplicative Haar) and $M3 \leftrightarrow M4$ (Gibbs detailed balance $\rightarrow$ antiperiodicity T_3) are explicitly constructed; the full link $M1 \leftrightarrow M2 \leftrightarrow M3 \leftrightarrow M4$ in the sense of a strict mathematical equivalence requires a categorical formulation not included here. The three roles of $s = 1/2$ listed in §2.1 (arithmetic **T1**, geometric **T5**, spectral Haar) are operator-theoretically identified by this section. This is the main structural contribution of the programme.

5 Cohomological Overdetermination of the Residue $1/8$

The semi-classical PT density differs from the Riemann–von Mangoldt density by a constant $1/8$ (§4.1). Berry and Keating 1999 identify this constant as the fine Maslov phase of the corners of the double barrier, without dynamical interpretation. PT provides a *triple cohomological identification* that converges, overdetermined by the coherence of the three readings.

Mechanism	Description	PT origin
M1 — Symplectic cell	$u_\star p_\star = 2\pi$	$[u, p_u] = i$ (Liouville)
M2 — Haar Jacobian	$du = dp/p$, shift $+1/2$	T2 , double stochasticity
M3 — Unitary transformation	$U = e^{-su/2}$	Gibbs detailed balance $\rightarrow$ SA
M4 — Boundary condition	$\theta = \pi$ antiperiodic	T_3 , antidiagonal of the sieve

Table 1: The four equivalent mechanisms for the critical line $\text{Re}(s) = 1/2$.

5.1 K2 — Algebraic identity: $1/8 = s^2/2 = \lambda_H$

The scalar Higgs self-coupling in the PT sector is derived in the monograph (ch. 15, depth $d = 3$) [6]:

$$\lambda_H = \frac{s^2}{2}. \quad (25)$$

At $s = 1/2$, $\lambda_H = 1/8$. This identity is structurally tied to the two-branch bifurcation q_+/q_- : λ_H is the square of the bifurcation parameter divided by the number of scalar branches.

Associated algebraic identifications. Two other algebraic identifications coincide:

$$\frac{1}{8} = \left(\frac{1}{2}\right)^3 \quad (\text{cube of the scalar unit, [4]}),$$

$$\frac{1}{8} = \frac{1}{2N_{\text{gen}} + 2} \quad (\text{at } N_{\text{gen}} = 3, \text{ exact}).$$

Rejected naive candidates. Several candidates are rejected by numerical incompatibility: $1/N_{\text{Weyl}} = 1/15 \neq 1/8$, $\chi(\Sigma_{\text{pers}})/Z = -26/Z \neq 1/8$, $\gamma_3\gamma_5\gamma_7 \approx 0.335 \neq 1/8$, $\alpha_{\text{bare}} \approx 1/136 \neq 1/8$. The $1/8$ is not a topological invariant of the persistence surface, but a scalar invariant of the bifurcation q_+/q_- .

5.2 K3 — Cohomological identity: $1/8 = c_1/N_{\text{corners}}$

The relevant cohomological structure is constructed in the PT corpus. The doubled Fisher manifold

$$\mathcal{M}_m^{\text{db}} = \mathcal{M}_m \times \mathcal{M}_m, \quad g = g_{\text{Fisher}} \oplus g_{\text{Fisher}}, \quad (26)$$

equipped with the complex structure $J_{\text{PT}}(u, v) = (-v, u)$ is a Kähler manifold (theorem `kahler_fisher`, [4]). The chiral projectors

$$P_{\pm} = \frac{1}{2}(I \mp iJ_{\text{PT}}) \quad (27)$$

decompose $H^*(BG_m; \mathbb{C})$ into $\pm i$ eigenspaces and are identified with the spinor projector q_+/q_- of the sieve ([4], `def:chiral_projectors`).

Theorem 5.1 (First Chern class of the PT spinor bundle, `berry_3pi`). *The first Chern class of the spinor bundle over $\mathcal{M}_m^{\text{db}}$ equals*

$$c_1 = s = \frac{1}{2}. \quad (28)$$

The holonomy of the N_{gen} -fold cover is $\exp(2\pi i c_1 N_{\text{gen}}) = \exp(3\pi i)$ (origin of the `berry_3pi` theorem of the monograph, app. P, § C7).

The algebraic sub-statement $c_1 = 1/2$ is **[Thm]** unconditional; full APS rigour of the calculation $1/8 = c_1/N_{\text{corners}}$ on the cornered manifold remains partially open (§9), status **[Der]/[partial]**.

Proposition 5.2 (Cohomological computation of the residue $1/8$). *In the canonical phase space (u, p_u) , the BK double barrier is a symplectic quarter-plane $Q_\star = \{u \geq u_\star, p_u \geq p_\star, u_\star p_\star = 2\pi\}$. The boundary admits two edges meeting at the corner. With the Weyl symmetrisation and the two mirror branches identified by the antiperiodicity T_3 (§4.4), one obtains*

$$N_{\text{corners}} = 4 \quad (29)$$

distinct corners on the complete cell. The Berry phase contributed by each corner, measured on the quarter-turn joining the two adjacent edges, is

$$\gamma_{\text{corner}} = -\frac{2\pi c_1}{N_{\text{corners}}} = -\frac{\pi}{4}, \quad (30)$$

and the associated Bohr–Sommerfeld density is

$$N_{\text{corner}} = \frac{\gamma_{\text{corner}}}{2\pi} = -\frac{c_1}{N_{\text{corners}}} = -\frac{1}{8}. \quad (31)$$

Sketch. The symplectic quarter-plane admits 2 edges ($u = u_\star$ and $p_u = p_\star$) that meet at the corner $(u_\star, p_\star) = (\sqrt{2\pi}, \sqrt{2\pi})$. The Weyl symmetrisation (involution $(u, p_u) \leftrightarrow (p_u, u)$) creates the mirror sector, doubling the edges and producing two additional corners. The antiperiodicity $\theta = \pi$ (proposition 4.3) identifies two pairs of corners into two, giving $N_{\text{corners}} = 4$ in total (without the identification, one would have $N = 8$ and $c_1/N = 1/16$, contradicting the observed residue (21)). The Berry phase per corner is computed by parallel transport on the inner quarter-turn: in local polar variables centred on the corner, $\gamma_{\text{corner}} = \oint A = -2\pi c_1/N_{\text{corners}} = -\pi/4$ with $c_1 = 1/2$. The passage to the Bohr–Sommerfeld density divides by 2π : $N_{\text{corner}} = -1/8$. Numerical verification: machine precision $1.68 \cdot 10^{-14}$ via the script `pt_k3_cohomological_maslov.py`. $\square$

The identification $1/8 = c_1/N_{\text{corners}}$ therefore yields the Berry–Keating $-\pi/8$ per corner, but *derived* from the PT cohomological structure, not as an opaque geometric artefact.

5.3 Berry phase per corner: $1/8 = s/4$

At the equator of the Bloch sphere ($\theta = \pi/2$), the Berry connection of a spin- s is constant: $A_\varphi = -\sin^2(\pi/4) = -1/2 = -s$ per unit azimuthal angle. Over a complete equatorial cycle, the Berry phase is $\gamma_1 = -\pi = -2\pi s$. Divided by the four corners (29):

$$\frac{\gamma_1}{N_{\text{corners}}} = -\frac{\pi}{4} = -2\pi \cdot \frac{s}{4}, \quad (32)$$

that is, $-1/8$ per corner in area count. The corresponding algebraic identity is

$$\frac{1}{8} = \frac{s}{4}. \quad (33)$$

5.4 Overdetermination by coherence

The three identifications converge to the same number at $s = 1/2$:

$$\frac{1}{8} = \underbrace{\frac{s^2}{2}}_{\text{K2, Higgs}} = \underbrace{\frac{c_1}{N_{\text{corners}}}}_{\text{K3, Chern}} = \underbrace{\frac{s}{4}}_{\text{Berry per corner}}. \quad (34)$$

The algebraic coherence between K2 and the Berry phase per corner reduces to

$$\frac{s^2}{2} = \frac{s}{4} \iff s(2s - 1) = 0. \quad (35)$$

The only solutions are $s = 0$ (trivial, no active parameter, excluded by **T1**) and $s = 1/2$ (theorem **T1**). **The coherence of the three cohomological readings overdetermines $s = 1/2$ modulo the assumed T1 axiom**; it does not derive $s = 1/2$ independently of **T1**. It is a structural signature of **T1** from the HP-PT zone of the corpus, not a substitute for **T1**.

5.5 Index-theorem reading

The final assembly is interpreted as a formal Atiyah–Patodi–Singer index theorem [9] on the cornered manifold $\square$:

$$\text{ind}(D_+|_{\square}) = \int_{\square} \text{ch}(E) \text{Td}(\square) + \frac{1}{2} (\eta_{\partial} + \dim \ker D_{\partial}) + \sum_{\text{coins}} \frac{\delta_{\text{coin}}}{2}. \quad (36)$$

With $c_1(E) = 1/2$ and $\delta_{\text{coin}} = \pi/2$ for a right corner, the per-corner contribution is $\pi/2/(2 \cdot 2\pi) = 1/8$, in agreement with the direct Berry calculation. The equivalent identification via Chern–Simons: $\exp(2\pi i c_1/N_{\text{corners}}) = \exp(i\pi/4)$, whose argument $\pi/4$ gives in density $1/8$ per corner.

Formal rigour in the strict APS sense (beyond the area calculation and numerical verification) remains to be developed (§9). The cohomological mechanism is nonetheless explicitly identified.

6 Higgs $\leftrightarrow$ Zeta Duality

The triple identification of §5 leads to an unexpected structural conjecture, formulated as verrou K4.

6.1 The common canonical bifurcator

Conjecture 6.1 (K4, Higgs $\leftrightarrow$ zeta duality). There exists a canonical PT object B , the *bifurcator*, such that:

- $m_H/v = \text{scale}(B) = s = 1/2$ (breathing mode, Higgs mass);
- $\Delta_N = \text{coupling}(B) = s^2/2 = 1/8$ (Maslov phase, HP-PT sector);
- $\lambda_H = \text{coupling}(B) = s^2/2 = 1/8$ (Higgs self-coupling).

The canonical bifurcator is identified as the spinor projector q_+/q_- of the channel $p = 2$ at the fixed point $\mu^* = 15$. Formally,

$$B = \Pi_+ : \mathcal{H} \rightarrow \mathcal{H}_{q_+}, \quad \Pi_+ = \frac{1}{2}(I - iJ_{\text{PT}}), \quad (37)$$

where J_{PT} is the Kähler–Fisher complex structure of $\mathcal{M}_m^{\text{db}}$ (cf. §5.2) and $\mathcal{H} = \mathcal{H}_{q_+} \oplus \mathcal{H}_{q_-}$ the chiral decomposition. The mirror operator $\Pi_- = (I + iJ_{\text{PT}})/2$ projects onto q_- , and B restricts to the channel $p = 2$ (eigenspace of the binary partition **T6**).

6.2 The $p = 2$ channel carries both observables

The $p = 2$ channel plays three roles in PT ([3]):

- info / anti-info operator ($\gamma_2 = 0.867$, the most active);
- bifurcation q_+/q_- (persistence chirality);
- binary partition $\log_2 m$ (theorem **T6**, GFT).

The Higgs lives in the channel $p = 2$ ([4]): it is neutral under $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$, the three gauges live in $\{3, 5, 7\}$, so only $p = 2$ is admissible for H .

The Maslov phase also lives in the channel $p = 2$: each corner of the BK double barrier is a point where the sieve switches $q_+ \leftrightarrow q_-$, which is the operational expression of the binary partition $p = 2$ (**T6**).

6.3 Explicit derivation by the chain **T1** + **T2** + Haar + Weyl

Theorem 6.2 (Higgs–Maslov identity via the canonical chain). *At $s = 1/2$,*

$$\boxed{\frac{1}{8} = \frac{s^2}{2} = \frac{|\lambda_2(T_{30})|}{N_{\text{branches}} q_+/q_-} = \frac{1/4}{2} = \lambda_H = \Delta N_{\text{Maslov}}.} \quad (38)$$

Sketch, four-step chain. **T1.** $s = 1/2$ (theorem, [3]).

T2. $|\lambda_2(T_{30})| = s^2 = 1/4$: the second eigenvalue of the primordial transfer matrix T_{30} equals s^2 (spectral gap, [3]).

Haar. Shift $\text{Re}(s_\zeta) \rightarrow \text{Re}(s_\zeta) + 1/2$ by the Jacobian of the multiplicative Haar measure (§4.2).

Weyl. Weyl symmetric ordering divides the state count by 2 (symmetrisation $(u, p_u) \leftrightarrow (p_u, u)$ of §4.1), which transforms s^2 into $s^2/2$.

The composition of the four steps gives $|\lambda_2|/2 = s^2/2 = 1/8$. Identifying the Higgs sector to $\lambda_H = s^2/2$ and the HP-PT sector to $\Delta_N^{\text{Maslov}} = s^2/2$, one obtains (38). $\square$

The quantity $1/8$ is therefore structurally the **T2** spectral gap divided by the number of q_+/q_- branches. The two readings (Higgs and Maslov) are two names for the same number, generated by the same mechanism.

6.4 Isolation tests

No other canonical PT coupling equals $1/8$:

PT coupling	Value	$= 1/8?$
α_{EM}	$7.30 \cdot 10^{-3}$	no
$\sin^2 \theta_W$	0.231	no
G_F (GeV^{-2})	$1.17 \cdot 10^{-5}$	no
λ_H	0.125	yes
ΔN_{Maslov}	0.125	yes

Table 2: Isolation tests: no PT coupling other than λ_H and ΔN_{Maslov} equals $1/8$.

Pure numerical coincidence is rejected. The spin hierarchy $(s, s^2, s^2/2)$ is dense up to $1/8$ and closes algebraically beyond ($s^3 = s^2/2$ at $s = 1/2$).

6.5 Quantitative numerical tests

At tree level, $m_H^{\text{tree}} = v \cdot s = 123.11$ GeV (using $v = 246.22$ GeV). At first PT perturbative order,

$$m_H^{(1)} = v \cdot s \cdot (1 + C_F \varepsilon) \simeq 124.65 \text{ GeV}, \quad (39)$$

then the full PT-NLO value (with higher-order corrections, [4]) reaches $m_H^{\text{NLO}} = 125.287$ GeV, where the two coefficients are defined intrinsically in PT:

- $C_F = (N_c^2 - 1)/(2N_c) = 4/3$ is the Casimir factor of the fundamental representation of $\text{SU}(N_c)$, with $N_c = 3$ imposed by PT as the only value of N_c for which the fixed-point equation $\mu^* = \sum p_i$ admits a solution on the active primes (theorem **T0**, [5]). Geometrically, C_F counts the allowed transitions of the transfer matrix T_3 of the $p = 3$ channel (7 transitions over 9 slots; [4]). Status **[Thm]** for the derivation $N_c = 3 \Rightarrow C_F = 4/3$.
- $\varepsilon = \alpha_s s/(2\pi)$ is the small PT parameter of the one-loop perturbative expansion, combining the strong coupling $\alpha_s(m_Z) \simeq 0.1179$ and the bifurcation parameter $s = 1/2$ with the standard $1/(2\pi)$ normalisation of a gluon loop ([4]). Numerically $\varepsilon \simeq 9.38 \cdot 10^{-3}$ and $C_F \varepsilon \simeq 1.25 \cdot 10^{-2}$. Status **[Der]** (standard perturbative formula with α_s and s taken at their canonical PT values, with no free parameter).

The LHC experimental value is 125.25 GeV. The first-order error $m_H^{(1)}$ is $\sim 0.5\%$, and the full PT-NLO error m_H^{NLO} is 0.0295 %.

6.6 Final position

K4 is established as a **structurally validated strong conjecture**. Pure numerical coincidence is rejected. The common mechanism is identified: q_+/q_- bifurcation of the $p = 2$ channel. The derivation chain $\mathbf{T1} \rightarrow \mathbf{T2} \rightarrow \text{Haar} \rightarrow \text{Weyl}$ uses only **[Thm]** theorems of the corpus. Promotion to an unconditional theorem requires the strict cohomological closure K3 (currently **[Thm partial]**, §9).

7 Numerical Verifications

This section summarises the main numerical verifications of the programme. Each line refers to the section that carries its derivation or description.

7.1 Summary table

Test	Precision / score	Reference
Fredholm identity $\det(I - D_R)^{-1} =$ $R(s), \operatorname{Re}(s) > 1$	$\sim 2 \cdot 10^{-12}$	§3.2
Trace formula $-d \log R/ds =$ $\sum_p (\log p) [1/(p^s - 1) + A_p p^{-s}]$	$\sim 10^{-26}$	thm. 3.5
Difference $N_{\text{PT}} - N_{\text{RvM}} = 1/8$ constant over 4 decades	$\sim 10^{-15}$	§4.1
Spectral symmetry $\{\gamma_n, -\gamma_n\}$ (AP BC, $\theta = \pi$)	$5 \cdot 10^{-12}$	prop. 4.3
Berry phase per corner $-\pi/4$, density $-1/8$	$1.68 \cdot 10^{-14}$	prop. 5.2
$\varepsilon_*^{\text{geom}}$ vs empirical optimum $W = 24$	$\sim 5\%$ global, $\sim 1\%$ at $M = 1155, W = 24$	lem. 3.4
Poles of $ T(1/2 + it) $ (raw formula) on $t \in [10, 70]$	17/17 at $ \Delta < 1.5$	§3.5
Poles of $ T(1/2 + it) $ (raw formula) on $t \in [10, 200]$	74/79 at $ \Delta < 1.5$ (sat. $P_{\text{max}} = 3000$)	§3.5
Poles of $ T(1/2 + it) $ with $R_1(\varepsilon = 0.2)$ on $t \in [10, 200]$	77/79 at $ \Delta < 1.5$, 75/79 at $ \Delta < 0.5$, 47/79 at $ \Delta < 0.1$	§3.5
Poles of $ T(1/2 + it) $ with $R_1(\varepsilon = 0.2)$ on $t \in [10, 500]$	267/269 (99.3%), 242/269 (90.0%) at $ \Delta < 0.5$, 139/269 (51.7%) at $ \Delta <$ 0.1	§7.4

Table 3: Main numerical verifications of the HP-PT programme.

7.2 Exact functional form of N_{PT}

Verification of (19) on $\gamma \in \{14, 21, 30, 50, 100, 200, 500, 1000\}$:

7.3 Distributional capture of zeros

Test on $t \in [10, 70]$ with Abel–Hadamard regularisation $p^{-\varepsilon}$. For $\varepsilon \in \{0.6, 0.4, 0.2\}$, the modulus $|F_\varepsilon(t)|$ systematically grows by a factor of $2.7\times$ to $3.3\times$ at the zeros of ζ when ε goes from 0.6

γ	N_{PT}	N_{RvM}	difference
14.13	0.5737	0.4487	0.1250
21.02	1.6945	1.5695	0.1250
50.00	9.5478	9.4228	0.1250
100	29.1273	29.0023	0.1250
500	269.7117	269.5867	0.1250
1 000	648.7412	648.6162	0.1250

Table 4: Difference $N_{\text{PT}} - N_{\text{RvM}}$ constant at $1/8$ over four decades, to machine precision ($\leq 10^{-15}$).

to 0.2 (signature of a pole), against $1.4\times$ to $2.3\times$ at neutral points (regularity). On $t \in [10, 70]$, $dt = 0.1$, we detect 41-52 local peaks depending on $P_{\text{max}} \in [3\,000, 100\,000]$; all 17 reference zeros are within 1.5 units of a peak ($|\Delta| < 1.5$) from $P_{\text{max}} = 3\,000$.

P_{max} saturation. Capture saturates beyond $P_{\text{max}} = 3\,000$: the median $|\Delta|$ decreases from 0.49 ($P_{\text{max}} = 3\,000$) to 0.36 ($P_{\text{max}} = 30\,000$) then stagnates. The same statistic extended to $t \in [10, 200]$ gives 74/79 zeros (94 %) at $|\Delta| < 1.5$, 38 at $|\Delta| < 0.5$, 7 at $|\Delta| < 0.1$. The 5 missed zeros belong to pairs (γ_n, γ_{n+1}) with gap below the intrinsic width of an unregularised peak ($\sim 2\pi/\log \gamma$), merged into a single peak. Extreme precision case: $\gamma = 146.001$ captured at $|\Delta| = 0.001$. Systematic sub-0.1 improvement requires applying explicit regularisation R_1 or R_2 , not increasing P_{max} .

7.4 Gaussian smearing regularisation and extended range

With $\phi_\varepsilon(t) = (\varepsilon\sqrt{2\pi})^{-1} \exp(-t^2/2\varepsilon^2)$ and $\varepsilon \in \{2, 1, 0.5\}$, the pole/neutral contrast is more pronounced: $8\times$ to $10\times$ at the zeros against $2\times$ to $3\times$ at neutral points. Gaussian and Abel regularisations are equivalent in the distributional sense (different regularising kernels), but the former is numerically more sensitive.

Extended range $t \in [10, 500]$ (269 zeros). R_1 with $\varepsilon = 0.2$ captures 267/269 zeros at $|\Delta| < 1.5$, 242/269 at $|\Delta| < 0.5$ and 139/269 at $|\Delta| < 0.1$. The breakdown by band of 100 units is:

Band	Zeros	Capture $ \Delta < 0.5$
[10, 100]	38	100.0 %
[100, 200]	41	92.0 %
[200, 300]	50	89.8 %
[300, 400]	67	87.5 %
[400, 500]	73	86.6 %

Table 5: Capture per 100-unit sub-band, R_1 Gaussian smearing $\varepsilon = 0.2$. Monotonic but bounded degradation: the performance does not collapse with T .

The floor at 86.6 % is consistent with the structural limit of 55 merged pairs (gap $< 2\pi/\log \gamma$) out of 269 zeros, i.e. 20.4 % of zeros belonging to a pair below the intrinsic model resolution.

ε -adaptive strategy tested. An ε -adaptive strategy ($\varepsilon \propto 1/\log \gamma$) was tested in two forms: convolution on raw $|F|$ (peak collapse, falsified) and convolution on complex F (marginal compromise: median $|\Delta|$ reduced to 0.080 but coverage reduced to 76 % at $|\Delta| < 1.5$). Fixed spectral-weight regularisation (R_1 , $\varepsilon = 0.2$) remains the **global optimum**.

8 Self-Correction: Dirichlet Prediction Falsified

The K4 duality (§6) suggests a natural extension to primitive Dirichlet L -functions via Haar factorisation mod q . This prediction was tested and falsified. This section honestly documents the test and draws from it the reformulation of the status of the residue $1/8$.

8.1 The prediction P3

If K4 is structurally extensible, the generic formula for the Maslov phase of $L(s, \chi \bmod q)$ would be

$$\Delta_N(L \bmod q) = s^2 \cdot \text{Haar}(\mathbb{Z}/q\mathbb{Z}) \cdot \frac{1}{N_{\text{branches}}} = \frac{1}{8q}. \quad (40)$$

Predictions: $q = 3 : 1/24$, $q = 5 : 1/40$, $q = 7 : 1/56$, $q = 11 : 1/88$.

8.2 Test M1 (mpmath, $T \leq 200$, 4 points per q)

First test, inconclusive: signal/noise too low (4 points, $\sigma_{\text{SE}} \sim 0.2$, target signal ~ 0.04).

8.3 Test M1bis (LMFDB, $T \leq 200$, ~ 140 zeros per q)

In-depth test on pre-computed LMFDB zeros for primitive quadratic characters $q \in \{3, 5, 7, 11\}$ (a parity of the character):

q	a	N zeros	$\langle \Delta_N \rangle$	SE	$1/(8q)$	$ t _{\text{PT}}$	$ t _0$
3	1	114	$-2.4 \cdot 10^{-4}$	0.019	+0.042	2.19	0.01
5	0	129	$-2.6 \cdot 10^{-3}$	0.021	+0.025	1.34	0.13
7	1	140	$+1.8 \cdot 10^{-3}$	0.021	+0.018	0.77	0.09
11	1	155	$-2.8 \cdot 10^{-4}$	0.021	+0.011	0.55	0.01

Table 6: Results of test M1bis. The t -stat compares the observed mean to the PT prediction $+1/(8q)$ (column $|t|_{\text{PT}}$) and to the null hypothesis $\Delta_N = 0$ (column $|t|_0$).

Discriminatory χ^2 test on 4 d.o.f.:

Hypothesis	χ^2	p -value
$H_0: \Delta_N = 0$	0.023	≈ 1.00
$H_{\text{PT}}: \Delta_N = +1/(8q)$	7.49	0.11
$H_{-\text{PT}}: \Delta_N = -1/(8q)$	7.05	0.13

Table 7: χ^2 test: H_0 is preferred by $\sim 300\times$ over H_{PT} . The prediction P3 is **effectively falsified** at $T \leq 200$.

Sign discussion. On the four values of q tested, PT predicts a strictly positive shift $+1/(8q)$; we observe three negative shifts ($q = 3, 5, 11$) and one positive ($q = 7$). The sign agreement alone is therefore $1/4$, well below the 1 that the prediction would impose. Falsification is not only amplitude-wise: the sign structure itself contradicts the prediction $+1/(8q) > 0$.

8.4 Calibration on ζ

With the same methodology on the first 50 zeros of ζ ([10]):

$$\langle \Delta_N \rangle_{\zeta} = +1.004 \pm 0.028. \quad (41)$$

Comparisons:

Target	Value	t
Pole of ζ alone (+1)	1.000	+0.15 (compatible)
Pole + Maslov ($1 + 1/8 = 1.125$)	1.125	-4.29 (rejected)
Maslov alone ($1/8$)	0.125	+31.29 (rejected)

Table 8: Calibration on ζ : the +1 from the pole entirely absorbs the $1/8$ in the counting statistic.

8.5 Reformulation of the status of the residue $1/8$

Critical consequence. Proposition 4.2 states that $N_{\text{PT}}(\gamma) - N_{\text{RvM}}(\gamma) = 1/8$, verified to 10^{-15} over four decades. This difference is between **two smooth asymptotic forms** (N_{PT} and N_{RvM}), not a measurable residue in the exact counting of zeros. At the count of real zeros, the $+1/8$ between PT and RvM is absorbed by the other terms in the formula and does not appear as a measurable mean shift. The +1 from the ζ pole entirely absorbs the $1/8$ in the counting statistic.

The asymptotic identity (21) remains valid as an **asymptotic identity between forms**, but the translation “ $1/8$ visible in the zero count” is erroneous. This explains a posteriori why the prediction $1/(8q)$ has been falsified: it attributed to Dirichlet a measurable residue analogous to that of ζ , while even for ζ this residue is not measurable in the zero count at heights $T \sim 200$.

8.6 Position in the programme

The falsification of P3 does not affect the HP-PT structure. Remain intact:

- the PT spectral model of ζ zeros via $T(s)$: 17/17 zeros captured at $|\Delta| < 1.5$ on $t \in [10, 70]$, 74/79 on $t \in [10, 200]$ with the raw formula, 75/79 at $|\Delta| < 0.5$ after explicit regularisation R_1 with $\varepsilon = 0.2$, and 267/269 at $|\Delta| < 1.5$ extended to $t \in [10, 500]$;
- the cohomological identification $1/8 = s^2/2 = c_1/N_{\text{corners}} = s/4$ for ζ ;
- the Higgs $\leftrightarrow \zeta$ duality via the $p = 2$ channel (conjecture 6.1) as a structural strong conjecture for ζ .

What falls: the natural extension to Dirichlet via the mod q Haar measure. The PT mechanism described here is **specific to ζ** , not to the Dirichlet family via this factorisation. The PT ledger marks P3 as **[FALSIFIED, $T \leq 200$]**.

This is the programme’s falsifiability discipline: a testable PT prediction was produced by K4, tested, rejected by data, and the programme restructures cleanly.

9 Limits and Openings

9.1 What is solidly established

The HP-PT programme, in its current state, provides:

- the closed-form expression of the amplitudes $\kappa_p(s)$ and the exact Fredholm representation of $R(s)$ (thm. 3.2);
- the canalized operator $\mathcal{H}_{\text{PT}}$ for the coupling with canonical $\varepsilon_*^{\text{geom}}$ without free parameter (lem. 3.4);
- the dynamical cut-off $u_{\text{max}}(\gamma) = \gamma/\sqrt{2\pi}$ forced by the symplectic Planck cell (prop. 4.1);
- the exact semi-classical density $N_{\text{PT}}(\gamma) = N_{\text{RvM}}(\gamma) + 1/8$ (identical functional form) (prop. 4.2);
- the defect indices (1, 1) and the antiperiodic boundary condition forced by T_3 (prop. 4.3);
- the exact analytic trace formula $-d \log R/ds$ (thm. 3.5);
- the operatorial regularised trace $T(s)$ with distributional poles at the zeros of R (§3.5);
- the triple cohomological identification of the residue $1/8$ ((34));
- the structural Higgs $\leftrightarrow \zeta$ duality (conjecture 6.1).

Summary table of epistemic statuses.

Result	Status	Source
$s = 1/2$ (T1)	[Thm]	[3]
$\det(I - D_R)^{-1} = R(s)$, $\text{Re}(s) > 1$	[Thm]	thm. 3.2
Trace formula $-d \log R/ds$ in $\text{Re}(s) > 1$	[Thm]	thm. 3.5
Defect indices $(1, 1)$ and AP BC $\theta = \pi$	[Thm]	prop. 4.3
$c_1 = 1/2$ on spinor bundle q_+/q_-	[Thm]	thm. 5.1
Density $N_{\text{PT}} = N_{\text{RvM}} + 1/8$ (asymptotic identity)	[Der]	prop. 4.2
$1/8 = c_1/N_{\text{corners}}$ (strict APS rigour)	[Thm partial]	prop. 5.2
$\zeta_+ \zeta_- \neq 0$ on critical line	[Conj]	rem. 3.3
Strict analytic continuation of $R(s)$ outside $\text{Re}(s) > 1$	[Open]	op. 1 below
Distributional poles of $T(s)$ on $\text{Re}(s) = 1/2$ only	[Open]	op. 2 below
	($\equiv$ RH)	
Canonical choice of u_* ($\sqrt{2\pi}$ vs $\log 3$)	[Open]	op. 4 below
Structural Higgs $\leftrightarrow \zeta$ duality (K4)	[Strong Conj]	conj. 6.1
$\Delta_N(L \bmod q) = 1/(8q)$ for Dirichlet (P3)	[Falsified, $T \leq 200$]	§8.3
$G_{\text{HP4.d}}$: direct spectrum $= \{\gamma_n\}$	[Falsified]	§3.5

Table 9: Epistemic statuses of the programme’s results.

9.2 What remains open

Four structural openings, plus an underlying non-vanishing hypothesis.

Opening 1 (G_HP3.a). Analytic continuation of $R(s)$ outside $\text{Re}(s) > 1$. The trace $T(s)$ is defined by series in $\text{Re}(s) > 1$ and then extended distributionally to $\text{Re}(s) = 1/2$; the intermediate region $1/2 < \text{Re}(s) < 1$ requires classical analysis of R . This is essentially the classical problem of ζ continuation.

Opening 2 (strict RH). Prove that **all** distributional poles of $T(s)$ lie on $\text{Re}(s) = 1/2$. The programme establishes that the structure favours this line (four equivalent mechanisms, §4), but does not prove the strict exclusion of off-line poles. This is essentially RH.

Opening 3 (strict APS K3). Formal Atiyah–Patodi–Singer rigour for the cornered manifold of the BK double barrier, beyond the area calculation and numerical verification. The cohomological mechanism is identified (§5.5), its strict formalisation remains to be developed.

Opening 4 (K1, canonical choice of u_*). The lower barrier u_* admits two natural candidates: $u_* = \sqrt{2\pi}$ (fixed point of the involution $u \leftrightarrow p_u$, chosen here) and $u_* = \log 3$ (first PT active prime, arithmetic choice). Both produce the same asymptotic density $N(\gamma)$ but different $O(1/\gamma)$ corrections. The canonical status of the choice between them remains open. The article retains $u_* = \sqrt{2\pi}$ by involution symmetry, without claiming to close K1.

Underlying [Conj] hypothesis: non-vanishing $\zeta_+ \zeta_- \neq 0$. The identification between zeros of R and zeros of ζ used throughout the article relies on this non-vanishing. It is verified numerically on the tested zones but not proved (rem. 3.3). A partial failure would introduce extra zeros (zeros of $\zeta_{\pm}$) which would translate into extra poles of $T(s)$; these possible zeros must be treated as conjecturally absent.

9.3 Canonical position of the programme

The HP-PT programme is a precise operatorial framework for the zeros of ζ . It provides:

1. a closed-form expression for the zeros via $\kappa_p(s)$ (intermediary D_R);
2. a canonical regularised trace formula $T(s)$;
3. four equivalent mechanisms for $\text{Re}(s) = 1/2$;
4. a continuous/discrete dictionary unified by persistence;
5. a structural explanation of the known numerical coincidences (2π , $7/8$, $1/8$, critical line).

It does not bring:

1. a strict proof of classical RH;
2. a shortcut to the distributional analysis of the regularised trace.

This distinction is maintained throughout the programme: one seeks to **understand** the zero structure, not to **prove** them on the critical line. The passage to formal proof requires fine Schwartz analysis on $T(s)$, which is no easier than the classical RH proof.

9.4 Comparison with Berry–Keating–Connes

Aspect	Berry–Keating–Connes	PT
Phase space	$(x, p) \in \mathbb{R}_+^2$	$(u, p_u), u = \log p$
Operator	$H = xp + 1/2$	$H_{\text{PT-BK}} = (u p_u + p_u u)/2$
Measure	$dx dp/2\pi$	$du dp_u/2\pi$
Cut-off	$x \geq \ell_x, p \geq \ell_p, \ell_x \ell_p = 2\pi$	$u_* p_* = 2\pi$ (PT Planck cell)
Regularisation	adelic (Connes)	atomic PT
Trace	$\text{Tr}(K) = \sum_p h(\gamma_p)$ (Weil)	$\text{Tr}_{\text{reg}} = \sum_p \kappa_p [\text{Euler} + \text{coupling}]$
Holonomic coupling	none (zeta only)	κ_p (bifurcation q_+/q_-)
2π	not structurally interpreted	$[u, p_u] = i$, canonical Liouville
$1/8$	corner Maslov phase (arithmetic)	$s^2/2 = c_1/4 = s/4$ (spinor cohomology)
Spin $s = 1/2$	postulated	overdetermined by $s^2/2 = s/4$ modulo T1

Table 10: Berry–Keating–Connes vs PT comparison: the main contribution of PT is intrinsic enrichment (no number is posited ad hoc).

Historical clarifications. The original operator of Berry and Keating (1999) is $H = xp$; the form $H = xp + 1/2$ appears in the later formalisation of Sierra (2008) and Sierra–Townsend, which regularises xp via a Planck cell and exhibits the symmetric ordering. The regularisation of Connes (1996, 1999) is of a different nature: it operates on the adelic space of idèle classes, by a non-commutative trace formula. The PT formulation is atomic on the primes: neither adelic nor continuous on $\mathbb{R}_+$.

9.5 Perspectives

Three natural extensions.

- **Other L -functions.** The Dirichlet prediction having been falsified, the extension to L -functions must be rethought. The cohomology of the spinor bundle under *twist* by χ completely modifies c_1 ; a proper cohomological reformulation is open.
- **Maslov phase in continuous Berry phase.** The hypothesis proposes that $1/(8q)$ exists at the level of the continuous Berry phase along the spectral flow, but does not manifest in the zero count. To be tested.

- **Strict APS rigour.** Develop the formal index theorem on the cornered manifold of the double barrier (strict K3). Not urgent for the operatorial framework, important for structural coherence.

10 Conclusion

The PT zeta programme delivers a canonical spectral model of the non-trivial zeros of the Riemann zeta function from a single internal structure: the persistence dynamics with its axiom $s = 1/2$. The final balance can be summarised in five points.

1. **The canonical operatorial object is unique.** The regularised trace

$$T(s) = \text{Tr}_{\text{reg}}[(s - iH_{\text{PT-BK}})^{-1} D_R(s)] = \sum_{p \in \mathcal{P}_\gamma} \kappa_p(s) (\log p) \left[\frac{1}{p^s - 1} + A_p p^{-s} \right]$$

is constructed from the amplitudes κ_p of the q_+/q_- bifurcation and the PT Berry–Keating dilation operator. No free parameter. Exact analytic identity in $\text{Re}(s) > 1$ to 10^{-26} , distributional poles on $\text{Re}(s) = 1/2$ at the zeros of $R(s) \equiv \zeta(s)/(\zeta_+ \zeta_-)$.

2. **The critical line $\text{Re}(s) = 1/2$ admits four equivalent readings:** symplectic Planck cell $u_\star p_\star = 2\pi$, multiplicative Haar Jacobian, unitary transformation $e^{su/2}$, antiperiodic boundary condition forced by T_3 . The three roles of $s = 1/2$ (**T1** arithmetic, **T5** geometric, Haar spectral) are operatorially identified.

3. **The cohomological residue $1/8$ is triply determined.**

$$\frac{1}{8} = \frac{s^2}{2} = \lambda_H = \frac{c_1}{N_{\text{corners}}} = \frac{s}{4}.$$

The coherence $s^2/2 = s/4$ admits $s = 0$ or $s = 1/2$; **T1** excludes $s = 0$, which overdetermines $s = 1/2$: this is a structural signature of **T1** by the HP-PT zone, not a substitute for **T1**.

4. **Structural Higgs $\leftrightarrow \zeta$ duality.** The canonical bifurcator B (spinor projector q_+/q_- of the channel $p = 2$ at the fixed point $\mu^* = 15$) produces simultaneously $m_H = v \cdot s$ and $\Delta_N = \lambda_H = s^2/2$ via the chain **T1** + **T2** + Haar + Weyl. No other PT coupling equals $1/8$: pure numerical coincidence is rejected.

5. **Falsifiability discipline.** The Dirichlet prediction $\Delta_N(L \bmod q) = 1/(8q)$, structurally induced by K4, was tested on LMFDB zeros of four primitive quadratic characters and falsified ($\chi^2(\text{PT}) = 7.49$ against $\chi^2(H_0) = 0.023$ on 4 d.o.f.). The residue $1/8$ is an asymptotic identity between smooth forms, not a measurable shift in the zero count. The PT mechanism described here is specific to ζ .

Canonical position. We have **understood** the zero structure of ζ ; we have **not proved** their strict localisation on the critical line. The ultimate lock ($G_{\text{HP3.a}}$: strict analytic continuation of R) is equivalent to RH in the classical sense and does not seem any easier than the latter.

The main enrichment over Berry–Keating–Connes is the intrinsic reading of the coincidences: the 2π is the canonical commutator, the $1/8$ is the scalar Higgs self-coupling of PT, and the spin $s = 1/2$ of the critical line is overdetermined by the cohomological coherence of the construction. **No ingredient is ad hoc.**

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